

CHAPTER 1 : INTRODUCTION & BASIC CONCEPTS

Section	Core Revision Focus
1.1	Thermodynamics, energy, first/second laws, approaches
1.2	Dimensions, units, dimensional homogeneity, conversions
1.3	Systems, control volumes, mass/energy transfer
1.4	Properties, intensive/extensive, specific properties, continuum
1.5	Density, specific volume, SG, specific weight
1.6	State, equilibrium, state postulate
1.7	Processes, paths, cycles, steady flow
1.8	Temperature, zeroth law, temperature scales
1.9	Pressure, pressure relations, hydrostatics, Pascal's law
1.10	Barometer, manometer, differential pressure
1.11	Systematic engineering problem solving

1.1 THERMODYNAMICS AND ENERGY

CORE CONCEPTS

- **Thermodynamics:** science of energy, energy transformations, and interactions between energy and matter.
- Energy may be viewed as the **ability to cause changes**.
- **Conservation of energy:** energy cannot be created or destroyed; only transformed.

$$\Delta E = E_{\text{in}} - E_{\text{out}}$$

Law	Main principle
First law	Energy is conserved; energy is a thermodynamic property.
Second law	Energy has both quantity and quality; actual processes proceed toward lower energy quality.

THERMODYNAMIC APPROACHES

Approach	Basis	Feature
Classical	Macroscopic behavior	Direct, relatively simple
Statistical	Average microscopic behavior	More involved

- **Classical thermodynamics:** uses measurable macroscopic quantities; primary engineering approach.
- **Statistical thermodynamics:** relates macroscopic behavior to microscopic particle behavior.

Key second-law idea: heat naturally transfers from higher temperature to lower temperature, with degradation of energy quality.

1.2 IMPORTANCE OF DIMENSIONS AND UNITS

DIMENSIONS AND UNITS

- **Dimensions:** characteristics used to describe physical quantities.
- **Units:** magnitudes assigned to dimensions.
- **Primary dimensions:** fundamental quantities such as mass, length, time, temperature.
- **Secondary dimensions:** derived from primary dimensions, e.g. velocity, energy, volume.

System	Description
SI	International System of Units; decimal-based
English/USCS	United States Customary System

SI FUNDAMENTAL UNITS

Dimension	Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Temperature	kelvin	K
Electric current	ampere	A
Amount of light	candela	cd
Amount of matter	mole	mol

SI PREFIXES

Factor	Prefix	Symbol	Factor	Prefix	Symbol
10^{24}	yotta	Y	10^{-1}	deci	d
10^{21}	zetta	Z	10^{-2}	centi	c
10^{18}	exa	E	10^{-3}	milli	m
10^{15}	peta	P	10^{-6}	micro	μ
10^{12}	tera	T	10^{-9}	nano	n
10^9	giga	G	10^{-12}	pico	p
10^6	mega	M	10^{-15}	femto	f
10^3	kilo	k	10^{-18}	atto	a
10^2	hecto	h	10^{-21}	zepto	z
10^1	deka	da	10^{-24}	yocto	y

UNIT CONVENTIONS

- Unit names are lowercase: newton, pascal, joule.

- Symbols derived from proper names are capitalized: N, Pa, J.
- Unit symbols are not pluralized and normally have no period.

DIMENSIONAL HOMOGENEITY

Every term in a valid engineering equation must have compatible dimensions/units.

- Different units cannot be added or subtracted.
- Dimensional analysis can detect equation/calculation errors.
- Dimensional homogeneity is **necessary but not sufficient** for physical correctness.

UNITY CONVERSION RATIOS

A ratio of equivalent quantities is exactly one:

$$\frac{1 \text{ N}}{1 \text{ kg m/s}^2} = 1$$

$$\frac{1 \text{ lbf}}{32.174 \text{ lbm ft/s}^2} = 1$$

- Insert unity ratios without changing physical value.
- Arrange ratios so unwanted units cancel.
- The inverse ratio is also equal to 1.
- Preferred for unit conversion rather than introducing g_c merely to force units to match.

1.3 SYSTEMS AND CONTROL VOLUMES

BASIC DEFINITIONS

Term	Definition
System	Quantity of matter or region in space selected for study
Surroundings	Everything outside the system
Boundary	Real/imaginary surface separating system and surroundings

Boundary may be **fixed/movable** and **real/imaginary**; mathematically it has zero thickness.

SYSTEM CLASSIFICATION

System	Mass Transfer	Energy Transfer
Closed system	No	Yes
Isolated system	No	No
Open system	Yes	Yes

- **Closed system/control mass**: fixed amount of mass; heat/work may cross boundary.
- **Isolated system**: special closed system with neither mass nor energy crossing.
- **Open system/control volume**: selected region in space through which mass and energy may cross.

For a closed system:

$$m = \text{constant}$$

CONTROL VOLUME

- Boundary of a control volume = **control surface**.
- Control surface may be real/imaginary and fixed/moving.
- Proper control-volume selection can greatly simplify analysis.
- Typical devices: compressors, turbines, nozzles, pumps, boilers.

Feature	Closed System	Control Volume
Matter	Fixed amount	May change
Mass crossing boundary	No	Yes
Energy crossing boundary	Yes	Yes
Boundary	Real/imaginary	Real/imaginary
Boundary motion	Fixed/movable	Fixed/movable
Typical example	Piston-cylinder	Turbine/nozzle

1.4 PROPERTIES OF A SYSTEM

INTENSIVE AND EXTENSIVE PROPERTIES

- **Property:** any characteristic of a system, e.g. P, T, V, m .
- **Intensive:** independent of system size/mass.
- **Extensive:** depends on system size/mass.

Type	Examples
Intensive	T, P , density
Extensive	m, V , total momentum

Division test:

- Intensive $\rightarrow$ value remains unchanged.
- Extensive $\rightarrow$ value changes in proportion to amount of matter.

SPECIFIC PROPERTIES

An extensive property per unit mass is a **specific property**; hence it is intensive.

$$v = \frac{V}{m} \quad e = \frac{E}{m}$$

where v = specific volume and e = specific total energy.

CONTINUUM

- Matter is treated as a continuous, homogeneous substance rather than discrete molecules.
- Allows properties to vary continuously and be treated as point functions.
- Valid when system characteristic dimension $\gg$ molecular mean free path.

For oxygen at 1 atm and 20°C:

$$d_{O_2} \approx 3 \times 10^{-10} \text{ m}, \quad \lambda \approx 6.3 \times 10^{-8} \text{ m}$$

At very high vacuum/elevation, λ may become comparable to system dimensions $\Rightarrow$ continuum assumption may fail.

1.5 DENSITY & SPECIFIC GRAVITY

DENSITY AND SPECIFIC VOLUME

$$\rho = \frac{m}{V}$$

$$v = \frac{V}{m} = \frac{1}{\rho}$$

$$\rho v = 1$$

Quantity	Definition	SI Unit
Density, ρ	m/V	kg/m^3
Specific volume, v	V/m	m^3/kg

For differential elements:

$$\rho = \frac{\delta m}{\delta V}$$

- Gas density generally $\uparrow$ with pressure and $\downarrow$ with temperature.
- Liquids/solids are essentially incompressible; density is much less pressure-sensitive.

SPECIFIC GRAVITY

$$SG = \frac{\rho}{\rho_{\text{H}_2\text{O}}}$$

For water at 4°C:

$$\rho_{\text{H}_2\text{O}} = 1000 \text{ kg/m}^3$$

SG is **dimensionless**.

$SG < 1$	Less dense than water
$SG = 1$	Same density as water
$SG > 1$	Denser than water

Useful relation:

$$1 \text{ g/cm}^3 = 1 \text{ kg/L} = 1000 \text{ kg/m}^3$$

Hence the numerical value of SG equals density in g/cm^3 or kg/L .

Example:

$$SG_{\text{Hg}} = 13.6 \Rightarrow \rho_{\text{Hg}} = 13600 \text{ kg/m}^3$$

SPECIFIC WEIGHT

$$\gamma_s = \rho g$$

SI unit:

$$[\gamma_s] = \text{N/m}^3$$

1.6 STATE AND EQUILIBRIUM

STATE AND EQUILIBRIUM

- **State:** condition of a system described by its property values.
- Change in any property $\Rightarrow$ different state.
- **Equilibrium:** state of balance with no unbalanced potentials/driving forces.
- Thermodynamic equilibrium requires all relevant equilibrium conditions simultaneously.

Equilibrium	Condition
Thermal	No temperature difference
Mechanical	No pressure changes with time at any point
Phase	Mass of each phase remains at equilibrium level
Chemical	Chemical composition does not change with time

STATE POSTULATE

For a **simple compressible system**:

State is completely specified by two independent intensive properties.

Simple compressible system: electrical, magnetic, gravitational, motion, and surface-tension effects are absent/negligible.

- T and v can specify the state.
- For single-phase systems, T and P are independent.
- For multiphase systems, T and P can be dependent:

$$T = f(P)$$

Thus, T and P alone are insufficient to specify a two-phase state.

If another significant effect exists, an additional property is required for each important effect.

1.7 PROCESSES AND CYCLES

PROCESS

- **Process:** change from one equilibrium state to another.
- **Path:** series of equilibrium states through which the system passes.
- A process is characterized by initial state, final state, path, and system-surroundings interactions.

QUASI-EQUILIBRIUM PROCESS

System remains infinitesimally close to equilibrium throughout the process.

- Approximated by a sufficiently slow process.
- Properties remain nearly uniform.
- Easier to analyze.
- Work-producing devices deliver maximum work under quasi-equilibrium operation.

For quasi-equilibrium processes, every point on the path represents an equilibrium state.

For nonquasi-equilibrium processes, a path for the system as a whole cannot be defined; it may be represented by a dashed line between initial and final states.

COMMON PROCESSES

Process	Constant Property
Isothermal	$T = \text{constant}$
Isobaric	$P = \text{constant}$
Isochoric/isometric	$v = \text{constant}$

CYCLE

Initial state = Final state

A cycle may contain several processes but returns the system to its initial state.

STEADY-FLOW PROCESS

Steady	No change with time
Uniform	No change with location

For a steady-flow control volume:

$$V = \text{constant}, \quad m = \text{constant}, \quad E = \text{constant}$$

At a fixed point, properties do not change with time, although they may vary from point to point.

Typical applications: turbines, pumps, boilers, and other continuously operating devices.

1.8 TEMPERATURE AND THE ZEROth LAW OF THERMODYNAMICS

THERMAL EQUILIBRIUM AND ZEROth LAW

- Heat transfers from higher temperature to lower temperature until temperatures become equal.
- Thermal equilibrium $\Rightarrow$ equal temperature.

Zeroth law:

A in thermal equilibrium with C, B in thermal equilibrium with C $\Rightarrow$ A and B are in thermal equilibrium.

This provides the basis for temperature measurement using a thermometer as the third body.

TEMPERATURE SCALES

Scale	Ice Point	Steam Point
Celsius	0°C	100°C
Fahrenheit	32°F	212°F

ABSOLUTE TEMPERATURE SCALES

- SI thermodynamic scale: **Kelvin**, K.
- English thermodynamic scale: **Rankine**, R.
- Kelvin is written without degree symbol: 300 K, not 300°K.

Absolute zero:

$$0 \text{ K} = -273.15^\circ\text{C}$$

IDEAL-GAS TEMPERATURE SCALE

At sufficiently low pressure and constant volume:

$$T = a + bP$$

At zero absolute pressure:

$$T = bP$$

The ideal-gas temperature scale agrees with the thermodynamic scale over its usable range, but fails at extreme temperatures because of effects such as condensation, dissociation, and ionization.

TEMPERATURE CONVERSION

$$T(\text{K}) = T(^{\circ}\text{C}) + 273.15$$

$$T(\text{R}) = T(^{\circ}\text{F}) + 459.67$$

$$T(R) = 1.8T(K)$$

$$T(^{\circ}F) = 1.8T(^{\circ}C) + 32$$

For engineering approximations, $273.15 \approx 273$ and $459.67 \approx 460$.

TEMPERATURE DIFFERENCES

$$\Delta T(K) = \Delta T(^{\circ}C)$$

$$\Delta T(R) = \Delta T(^{\circ}F)$$

$$1 K = 1^{\circ}C = 1.8 R = 1.8^{\circ}F$$

Critical distinction:

- Temperature **itself** in thermodynamic equations $\rightarrow$ use absolute scale.
- Temperature **difference** $\rightarrow$ K and $^{\circ}C$ have identical magnitude.

1.9 PRESSURE

DEFINITION AND UNITS

$$P = \frac{F_n}{A}$$

Pressure is the normal force per unit area and is a scalar quantity.

$$1 \text{ Pa} = 1 \text{ N/m}^2$$

1 kPa	10^3 Pa
1 MPa	10^6 Pa
1 bar	$10^5 \text{ Pa} = 100 \text{ kPa}$
1 atm	$101325 \text{ Pa} = 101.325 \text{ kPa}$
1 kgf/cm ²	$9.807 \times 10^4 \text{ Pa}$
1 psi	1 lbf/in^2

Useful:

$$1 \text{ atm} = 1.01325 \text{ bar} = 14.696 \text{ psi}$$

$$1 \text{ kgf/cm}^2 = 0.9807 \text{ bar} = 0.9679 \text{ atm} = 14.223 \text{ psi}$$

ABSOLUTE, GAGE, AND VACUUM PRESSURE

$$P_{\text{gage}} = P_{\text{abs}} - P_{\text{atm}}$$

$$P_{\text{vac}} = P_{\text{atm}} - P_{\text{abs}}$$

Therefore:

$$P_{\text{abs}} = P_{\text{atm}} + P_{\text{gage}}$$

$$P_{\text{abs}} = P_{\text{atm}} - P_{\text{vac}}$$

Pressure	Reference	Meaning
Absolute	Vacuum	Actual pressure
Gage	Atmosphere	Pressure relative to atmosphere
Vacuum	Atmosphere	Amount below atmospheric pressure

Unless specified otherwise, thermodynamic pressure means absolute pressure.

HYDROSTATIC PRESSURE

For a fluid at rest:

- Same horizontal level in the same continuous fluid $\Rightarrow$ same pressure.
- Pressure increases with depth.

- Pressure acts normal to a surface.

For constant density:

$$\Delta P = P_2 - P_1 = -\rho g \Delta z$$

Equivalently:

$$P_{\text{below}} = P_{\text{above}} + \rho g |\Delta z|$$

or

$$P_{\text{below}} = P_{\text{above}} + \gamma_s |\Delta z|$$

For variable density:

$$\frac{dP}{dz} = -\rho g$$

$$P_2 - P_1 = - \int_1^2 \rho g \, dz$$

For gases, pressure variation with elevation is usually negligible over small/moderate elevation changes.

PASCAL'S LAW

Pressure applied to a confined fluid increases by the same amount throughout the fluid.

For connected pistons at the same level:

$$P_1 = P_2$$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$\frac{F_2}{F_1} = \frac{A_2}{A_1}$$

$$\text{IMA} = \frac{A_2}{A_1}$$

Larger piston $\Rightarrow$ larger output force.

1.10 PRESSURE MEASUREMENT DEVICES

BAROMETER

Measures atmospheric pressure.

$$P_{\text{atm}} = \rho g h$$

where h is mercury-column height.

- Column height is independent of tube length and cross-sectional area when capillary effects are negligible.
- Atmospheric pressure decreases with elevation.

Standard atmosphere:

$$1 \text{ atm} = 760 \text{ mmHg} = 760 \text{ torr}$$

$$1 \text{ torr} \approx 133.3 \text{ Pa}$$

A water column of approximately 10.3 m produces standard atmospheric pressure.

MANOMETER

A manometer determines pressure differences from liquid-column height.

Basic relation:

$$\Delta P = \rho g h$$

Movement	Pressure change
Downward through fluid	$+\rho g h$
Upward through fluid	$-\rho g h$
Horizontal in same continuous fluid	0

For a manometer open to atmosphere:

$$P_2 = P_{\text{atm}} + \rho g h$$

Inclined manometer: slanted tube increases liquid displacement for a given pressure difference $\Rightarrow$ higher resolution for small pressure differences.

MULTIFLUID MANOMETERS

For stacked immiscible fluids:

$$P_{\text{bottom}} = P_{\text{top}} + \rho_1 g h_1 + \rho_2 g h_2 + \rho_3 g h_3$$

If all densities are equal:

$$P_{\text{bottom}} = P_{\text{top}} + \rho g (h_1 + h_2 + h_3)$$

Analysis rules:

1. Downward $\rightarrow$ add ρgh .
2. Upward $\rightarrow$ subtract ρgh .
3. Same elevation in same continuous static fluid $\rightarrow$ equal pressure.

DIFFERENTIAL MANOMETER

Measures pressure difference between two specified points.

For the arrangement considered:

$$P_1 - P_2 = (\rho_2 - \rho_1)gh$$

where ρ_1 = flowing-fluid density, ρ_2 = manometer-fluid density.

For gases, $\rho_1 \ll \rho_2$:

$$P_1 - P_2 \approx \rho_2 gh$$

1.11 PROBLEM-SOLVING TECHNIQUE

SEVEN-STEP PROCEDURE

Step	Action
1	Problem statement: identify given data, unknowns, objective.
2	Schematic: draw system; show dimensions, properties, interactions, unknowns.
3	Assumptions: state and justify approximations.
4	Physical laws: identify system/control volume and apply relevant laws.
5	Properties: obtain required properties from relations/tables.
6	Calculations: substitute, solve, check units and significant digits.
7	Verification: check physical reasonableness, assumptions, validity, limitations.

EXAM/PROBLEM-SOLVING CHECKLIST

- Define the system/control volume before applying a law.
- State assumptions explicitly.
- Keep units throughout calculations.
- Check dimensional consistency.
- Retain sufficient intermediate digits.
- Check whether the final result is physically reasonable.
- Do not apply results outside the assumptions under which they were obtained.

Engineering solutions: neatness, organization, completeness, and clarity help reveal errors. Not every simple problem requires every step to be written explicitly.

SIGNIFICANT DIGITS

CORE RULES

- Significant digits represent meaningful precision in measured data.
- Final answer cannot be more precise than the least precise input.
- Keep full calculator precision during intermediate calculations.
- Round only the final answer.
- More digits do **not** imply greater accuracy.
- Experimental uncertainty limits meaningful precision.

Example:

$$V = 3.75 \text{ L}, \quad \rho = 0.845 \text{ kg/L}$$

$$m = V\rho = 3.75(0.845) = 3.16875 \text{ kg}$$

Since both inputs have three significant digits:

$$m = 3.17 \text{ kg}$$

ROUNDING UNCERTAINTY

For a value rounded to a given decimal place, maximum rounding uncertainty is half the value of the least significant place.

For 3.75:

$$\text{uncertainty} = \pm \frac{0.01}{2} = \pm 0.005$$

Therefore:

$$3.745 \leq x < 3.755$$

General rule:

$$\text{rounding uncertainty} = \pm \frac{\text{value of least significant digit}}{2}$$

ENGINEERING APPROXIMATIONS

Small intentional approximations may be acceptable when their error is negligible relative to other uncertainties.

Example:

$$\rho_{\text{water}, 75^\circ\text{C}} \approx 975 \text{ kg/m}^3 \rightarrow 1000 \text{ kg/m}^3$$

Approximate error:

$$\frac{1000 - 975}{975} \times 100\% \approx 2.5\%$$

Thus, reasonable rounding and approximation are acceptable when justified by the required accuracy.